# SUNLIGHT TRACKING SYSTEM USING ARDUINO UNO

**24ECP36 - DESIGN STUDIO - I**

**A MINI PROJECT REPORT**

**Submitted by**


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***In partial fulfilment of the award of the degree***

**of**

**BACHELOR OF ENGINEERING**

**in**

**ELECTRICAL AND ELECTRONICS ENGINEERING**


**K.S.R. COLLEGE OF ENGINEERING**

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**ANNA UNIVERSITY : : CHENNAI 600025**


**DECEMBER 2025**

**K.S.R. COLLEGE OF ENGINEERING**

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## BONAFIDE CERTIFICATE

Certified that this project report "**SUNLIGHT TRACKING SYSTEM USING ARDUINO UNO"** is the bonafide work of **CHANDRASEKAR S (73152414013), KAVIN (73152414048), DINESH KUMAR L (L5)** who carried out the project under my supervision**.**

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## ACKNOWLEDGMENT

We would like to place our sincere thanks & gratitude to our Founder **Theivathiru Lion Dr. K.S. RANGASAMY, MJF**, K.S.R. Educational Institutions & **Thiru R. SRINIVASAN B.B.M., MISTE.,** Chairman, K.S.R. Educational Institutions for providing us the needed facilities to do this project.

We express our heartfelt gratitude to respected Dean, **Dr.M.VENKATESAN, M.E., Ph.D.,** for their constant encouragement & support throughout the project.

We express our heartfelt gratitude to respected Principal, **Dr. P. MEENAKSHI DEVI, M.E., Ph.D.,** the woman behind the formation of thousands of professionals, for her valuable guidance & for motivating us to succeed in life.

We like to express our sincere thanks to respected Head of the Department of Electrical & Electronics Engineering**, Dr. S. RAMESH, M.E., Ph.D.,** for the guidance & support.

We express our sincere thanks to respected project coordinator, **Dr.M. VIJAYAKUMAR M.E., Ph.D.,** for supporting us in carrying out the project.

We express our sincere thanks to respected project guide, **Mr.M. SUBRAMANI, M.E.,** for his invaluable guidance & technical advice.

We would like to express our sincere gratitude to all the faculty & staff members of the Electrical & Electronics Engineering Department for their valuable support towards the completion of this project.

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## DECLARATION

We affirm that the project report titled **"SUNLIGHT TRACKING SYSTEM USING ARDUINO UNO"** submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering is the original work carried out by us. It has not formed part of any other project report or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidates.

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To verify that the declaration made by the above candidates is true to the best of my knowledge.

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# ABSTRACT

The increasing global demand for renewable and sustainable energy has brought solar power into the spotlight as a clean, reliable, and inexhaustible source of energy. However, the energy conversion efficiency of solar panels largely depends on their orientation toward the sun. A fixed solar panel captures maximum sunlight only during a specific time of day when the sun's rays are perpendicular to its surface.To overcome this limitation, the Sunlight Tracking System automatically adjusts the orientation of the solar panel in accordance with the movement of the sun. The system uses LDR (Light Dependent Resistor) sensors to detect sunlight intensity, an to process signals, and a servo motor to move the solar panel. This automatic tracking ensures that the panel always faces the sun, thereby increasing the efficiency of power generation by up to 40%. The system is designed to be low-cost, energy-efficient, and scalable for various solar applications.This dynamic adjustment ensures that the panel remains perpendicular to sunlight from sunrise to sunset, increasing power output and improving overall system efficiency. The design is compact, cost-effective, and suitable for small-scale renewable energy applications. Furthermore, it demonstrates the role of embedded systems, automation, and sensor-based control in modern energy solutions. This dynamic adjustment ensures that the panel remains perpendicular to sunlight from sunrise to sunset, increasing power output and improving overall system efficiency The project highlights how solar tracking technology can contribute to sustainable.

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| | |
|---|---|
| LDR | Light Dependent Resistor |
| DC | Direct Current |
| LED | Light Emitting Diode |
| VCC | Voltage Common Collector |
| GND | Ground |
| IOT | Internet of Things |
| USB | Universal Serial Bus |
| RPM | Rotations Per Minute |

# CHAPTER 1

## 1.1 INTRODUCTION

Energy is one of the most essential needs of modern society, and among all renewable energy sources, solar energy is considered the most abundant, clean, and reliable. Solar panels convert sunlight into electricity, but their efficiency depends on how much sunlight falls directly on the surface. In a fixed solar panel system, the panels remain in one position throughout the day, resulting in less energy production because the sun's position keeps changing from morning to evening.

To solve this problem, a Sunlight Tracking System is developed. This system automatically rotates the solar panel toward the sun's direction using sensors and a microcontroller. By continuously tracking the sun, the panel receives maximum sunlight throughout the day, which increases power generation by 25–35% compared to fixed solar panels.

This project uses components such as Arduino/NodeMCU, Light Dependent Resistors (LDRs) to detect sunlight, servo or DC motors to move the panel, and a power supply module. The system is economical, efficient, and suitable for homes, streetlights, agriculture, and small solar projects.

## 1.2 MOTIVATION FOR PROJECT

In today's world, the demand for energy is continuously increasing, while non-renewable energy sources like coal, petroleum, and natural gas are rapidly decreasing. This creates a need for clean and renewable energy sources. Solar energy is one of the best alternatives because it is free, pollution-free, and available in large amounts.

However, the major drawback of normal solar panels is that they remain fixed in one direction. As the sun moves from east to west during the day, the amount of sunlight falling on the panel reduces, resulting in less electricity generation. Almost 30–40% of solar energy is wasted due to this fixed-position setup.

To overcome this issue, the idea of a Sunlight Tracking System was developed. This system follows the sun's movement and keeps the solar panel always facing the sun. This increases the panel's efficiency and produces more electricity without increasing the size or number of panels.

This project is motivated by three main goals:

- To increase solar energy output without extra cost of more solar panels.
- To promote green energy and reduce environmental pollution.
- To design a system that is low-cost, automatic, and useful for rural and urban areas.

## 1.3 OBJECTIVE OF PROJECT

To design and develop an automatic system that tracks the movement of the sun. To improve the efficiency of solar panels using real-time tracking. To minimize energy loss caused by fixed positioning of panels. To create a cost-effective and easily buildable prototype using basic electronic components.

## 1.4 THEME

Our entire system operation is classified into two , first is the hardware design and operation and second is the software installation and working. At first, we aim at developing a system with hardware components such as power supply, LDR, lights and Arduino. After developing the kit, we aim to install software named MQQT Client and connect the mobile with Node MCU in the kit. The program is written in C language.

We used LED instead of fluorescent, CFL, high pressure sodium lamps or metal halide bulb. LEDs are energy efficient because they emit very little heat, so they require far less power to emit the same amount of light as their conventional equivalents. In comparison, incandescent bulbs release 90% of their energy as heat and fluorescent lights release about 80% of their energy as heat. LED lights are 40 to 60% more energy efficient than traditional lighting technologies.

LED lighting products last longer than conventional lights. For example, LED bulbs can last three to five times longer than fluorescent lights and 30 times longer than incandescent ones.

The high colour rendering of LED street lights is close to natural light and presents colours more realistically. This helps drivers and pedestrians better identify objects at night, improving traffic safety and road conditions. Incandescent lamps take about 0.1 to 0.2 seconds to light up. Gas discharge lamps such as high-pressure sodium lamps and metal halide lamps take anywhere from tens of seconds to 10 minutes to reach a stable light output. After being and they can work normally in a continuous on/off state

## 1.5 ORGANIZATION OF THE PROJECT REPORT

**Chapter 1** This chapter introduces the sunlight tracking system and explains why the project is important. It describes how fixed solar panels cannot follow the sun and therefore produce less energy.

**Chapter 2** The empathy chapter focuses on understanding the needs of users and real-world problems related to solar energy**.**

**Chapter 3** This chapter clearly defines the main problem and the goal of the project. The problem statement explains that fixed solar panels do not capture maximum sunlight throughout the day.

**Chapter 4** In this chapter, different ideas and possible solutions are generated to solve the defined problem. Brainstorming is done to explore various tracking mechanisms, sensor types, and designs

**Chapter 5** This chapter explains the prototype development and testing results. It describes how the sunlight tracking system was built using Arduino, LDRs, and servo motors.

## 1.6 SUMMARY

The Sunlight Tracking System is an automatic device that keeps a solar panel aligned with the sun throughout the day. It uses LDR sensors to detect the direction of maximum sunlight. When the light intensity changes, the microcontroller compares the sensor values and rotates the panel using a servo motor. This ensures that the panel always receives maximum sunlight, which increases power generation efficiency.

# CHAPTER 2

# EMPATHY

## 2.1 INTRODUCTION

Empathy is the ability to understand and share the feelings of another person. It allows us to imagine what someone else is experiencing and respond with care, kindness, or support. Empathy helps build strong relationships, improves communication, and creates a positive and supportive environment. It is an important emotional skill that promotes understanding, reduces conflict, and strengthens human connection.

## 2.2 UNDERSTANDING USER EMOTIONS

Understanding user emotions is a crucial step in creating meaningful and effective solutions. For the Sunlight Tracking System, it's essential to empathize with users such as homeowners, farmers, or solar panel operators who seek to maximize energy efficiency. Emotions like frustration due to power wastage or satisfaction from efficient energy use can guide the design process. By observing and interviewing users, we can identify their emotional responses to current solar systems—whether they feel helpless when systems fail to track sunlight properly or confident when energy output improves. Recognizing these emotional drivers helps us design a solution that not only functions efficiently but also creates a sense of trust, satisfaction, and empowerment among users.

## 2.3 USER-FRIENDLY DESIGN

A user-friendly design focuses on simplicity, convenience, and efficiency. The Sunlight Tracking System must be easy to install, operate, and maintain. This involves creating an intuitive interface, providing clear indicators for operation status, and ensuring that users can easily adjust or monitor the system without technical expertise. A simple display showing energy generation and panel orientation can make the design more interactive. The mechanical and software components should be optimized for reliability and minimal maintenance. A user-friendly approach enhances user engagement and encourages wider adoption of solar tracking technology.

## 2.4 ACCESSIBLE COMMUNICATION

Accessible communication ensures that information about the Sunlight Tracking System is understandable to all types of users, regardless of their technical background. This includes using simple language in manuals, offering visual aids or demonstrations, and possibly integrating mobile alerts or app-based notifications about system performance. Clear communication also involves providing real-time feedback on system status, such as sunlight intensity and panel angle, which helps users make informed decisions. Accessibility in communication strengthens trust, reduces confusion, and promotes the long-term usability of the system.

## 2.5 CUSTOMIZATION FOR INDIVIDUAL NEEDS

Customization allows the Sunlight Tracking System to meet diverse user requirements. Different users—such as residential homeowners, agricultural users, or industrial operators—may have varying needs regarding energy capacity, tracking precision, or panel size. Providing options for manual and automatic modes, adjustable sensitivity to sunlight, and compatibility with different solar panel models can make the system more versatile. Allowing users to set personalized parameters, such as energy output alerts or specific operational schedules, enhances their sense of control. Customization ensures that the system aligns with each user's unique context, leading to better satisfaction and improved usability.

## 2.6 REDUCING FALSE ALARMS

Reducing false alarms is vital to maintaining user trust and system reliability. In a sunlight tracking system, false alerts—such as unnecessary repositioning or error signals caused by brief cloud cover or sensor noise—can cause confusion and reduce efficiency. Implementing advanced algorithms, sensor calibration, and noise filtering techniques can minimize these issues. Regular self-diagnostic checks and clear alert differentiation (e.g., color-coded signals for minor vs. critical alerts) further improve accuracy. By minimizing false alarms, users feel more confident in the system's performance and are less likely to experience frustration or doubt.

## 2.7 EMOTIONAL REASSURANCE

When system malfunctions or power interruptions occur, users may experience stress or uncertainty. Providing emotional reassurance through design and communication can rebuild user confidence. This can include automated recovery messages, quick troubleshooting guides, or support notifications that inform users of corrective actions. Visual indicators like "System Restoring" or "Reconnected Successfully" help reassure users that the problem is under control. Additionally, offering easy access to customer support or maintenance assistance provides emotional comfort. Emotional reassurance after incidents transforms negative experiences into opportunities for trust and satisfaction.

## 2.8 SUMMARY

Empathy is the ability to understand another person's feelings and see things from their point of view. It helps people connect better, communicate effectively, and support each other. By showing empathy, we create stronger relationships, reduce misunderstandings, and build a more caring and compassionate environment.

# CHAPTER 3

# DEFINE STAGE

## 3.1 INTRODUCTION

The Define Stage is the second step in the Design Thinking process, where the information collected during the Empathy stage is analyzed and clearly understood. In this stage, the main goal is to identify the real problem that needs to be solved. By organizing insights, understanding user needs, and framing a clear problem statement, the Define stage helps set a focused direction for creating effective and meaningful solutions.

## 3.2 UNDERSTANDING THE DEFINE STAGE

The Define Stage is the second phase of the Design Thinking process. It focuses on analyzing the information gathered during the Empathize stage to clearly articulate the core problem that needs solving. In this phase, user observations, emotions, and needs are synthesized to form meaningful insights. For the Sunlight Tracking System, this means converting user frustrations—such as inefficient energy capture or high maintenance—into clear design challenges. Defining the problem accurately helps direct the solution toward real user requirements and ensures that innovation has a strong purpose.

## 3.3 PROBLEM IDENTIFICATION

Problem identification involves pinpointing the key challenges users face. Through interviews and observations, we can identify problems such as solar panels not facing optimal sunlight angles, energy loss due to manual adjustments, and difficulties in maintaining consistent performance. Many users express dissatisfaction with static panels that fail to capture sunlight throughout the day. Identifying these pain points allows us to focus on creating an automated system that optimizes panel orientation, increases energy output, and reduces human effort.

## 3.4 USER NEEDS AND INSIGHTS

User needs refer to what the users truly require from a solution, while insights represent deeper understandings of their motivations and pain points. In this case, users need a solar tracking system that is automatic, cost-effective, reliable, and low-maintenance. Insights reveal that users want peace of mind—knowing their system is working efficiently without constant supervision. These insights guide design decisions toward creating a smarter, more user-centered product that improves both functionality and user satisfaction.

## 3.5 CREATING A POINT OF VIEW STATEMENT

A Point of View (POV) statement captures the essence of the problem from the user’s perspective. It connects the user, their need, and the underlying insight.This statement helps maintain focus on the real problem throughout the design and prototyping process.

## 3.6 PROBLEM FRAMING

Problem framing involves structuring the identified issue in a way that encourages innovation and practical solutions. Instead of asking, “How can we make solar panels more efficient?”, we reframe it as, “How might we design a smart system that automatically tracks sunlight to maximize efficiency and user convenience?” Framing the problem this way opens creative possibilities and directs the team toward human-centered innovation rather than purely technical improvements.

## 3.7 DEFINE SUCCESS CRITERIA

Success criteria provide measurable indicators to evaluate whether the solution effectively meets user needs. For the Sunlight Tracking System, success can be defined by:

- Increased energy efficiency (e.g., 20–30% improvement in output)
- User satisfaction (positive feedback and ease of use)
- System reliability (minimal errors or downtime)

- Affordability and low maintenance
- Sustainability impact (reduction in energy waste)

## 3.8 SUMMARY

The Define Stage focuses on analyzing the information gathered during the Empathy stage and identifying the core problem to solve. It helps organize user needs, highlight key insights, and create a clear problem statement. This stage gives direction to the next steps in the design process by ensuring the team works on the right problem.

# CHAPTER - 4

# IDEATION

## 4.1 INTRODUCTION

The Ideation Stage is the third step in the Design Thinking process, where creative ideas and solutions are generated to address the problem defined in the previous stage. In this stage, brainstorming, sketching, and other creative techniques are used to think freely and explore multiple possibilities without judgment. The goal is to come up with innovative, practical, and user-centered solutions that can be developed further in the next stages

## 4.2 BRAINSTORMING INNOVATIVE ALARM

Brainstorming in the Ideation Stage encourages creativity and open thinking to explore multiple possible solutions. For the Sunlight Tracking System, innovative alarm concepts can enhance system performance and user safety. For example, an alert mechanism can notify users of system malfunctions, panel misalignment, or sensor blockage. These alarms can be visual (LED indicators), auditory (buzzers), or digital (mobile notifications). Brainstorming focuses on finding new ways to combine technology and usability—such as integrating solar-powered alarms or smart alerts that automatically reset after resolution—to keep users informed without unnecessary complexity.

## 4.3 SMART INTEGRATION IDEAS

Smart integration ensures the Sunlight Tracking System connects seamlessly with other smart technologies. The idea is to create a smart ecosystem where the tracking system can communicate with home automation systems or renewable energy monitoring platforms. For example, the system can be linked to a mobile app or IoT dashboard that displays real-time energy production, sunlight levels, and panel orientation. Integrating with cloud storage allows for performance tracking over time. This level of connectivity increases transparency, helps users make informed decisions, and modernizes the system for future scalability.

## 4.4 USER-FRIENDLY INTERFACE DESIGN

An effective user interface (UI) transforms a technical system into an approachable, easy-to-use product. The interface for the Sunlight Tracking System should include simple controls, clear data visualization, and intuitive navigation. For example, a touchscreen LCD or mobile app could show sunlight intensity, panel angle, and total energy generated. Using color-coded indicators (green for optimal, red for malfunction) simplifies user understanding. The design should cater to all users, including those with limited technical knowledge. A clear and accessible interface fosters trust, usability, and satisfaction.

## 4.5 MULTIPLE SENSOR OPTIONS

Incorporating multiple sensors increases the accuracy and efficiency of sunlight tracking. The system can use light-dependent resistors (LDRs), infrared sensors, or photo-diodes to detect sunlight intensity from various angles. Multiple sensors allow the system to adapt dynamically to environmental conditions such as cloud cover or partial shading. Using redundancy ensures reliability—if one sensor fails, others maintain performance. This feature makes the system robust, efficient, and adaptable to different climates and regions.

## 4.6 EMERGENCY RESPONSE FEATURES

Emergency response features add a layer of safety and reassurance. The Sunlight Tracking System can include auto-lock mechanisms to prevent damage during high winds, storms, or overheating. The system can also send alerts to users when panels face extreme conditions, ensuring timely preventive action. Integration with emergency protocols—such as automatic shutdown or safe mode activation—protects both the equipment and user investment. These features build trust and demonstrate the system's intelligence and reliability under unexpected conditions.

## 4.7 COST-EFFECTIVE DESIGN

A successful innovation balances functionality with affordability. The cost-effective design of the Sunlight Tracking System focuses on using readily available, low-cost components such as Arduino Uno, servo motors, and basic LDR sensors without compromising efficiency. By optimizing energy use, simplifying mechanical design, and reducing maintenance costs, the system provides long-term economic benefits. Modular construction also allows users to upgrade parts rather than replace the entire system

## 4.8 SUMMARY

The Ideation Stage is where creative solutions are generated for the defined problem. It involves brainstorming and exploring multiple ideas without judgment. The goal is to find innovative and practical solutions that can be developed in the next stages of the design process.

# CHAPTER – 5

# PROTOTYPE AND RESULTS

## 5.1 INTRODUCTION

The Prototype Stage is the phase in the design process where the ideas generated during Ideation are turned into tangible models or small-scale versions of the solution. Prototypes help test concepts, identify problems, and gather feedback from users. The Results part involves analyzing the performance of the prototype, evaluating whether it meets user needs, and making improvements. This stage is essential for refining the solution before final implementation.

## 5.2 COMPONENTS REQUIRED

To build an efficient Sunlight Tracking System, the following components are required:

### 5.2.1 Arduino Uno

Arduino Uno is a popular microcontroller board based on the ATmega328P microchip. It is widely used for building electronic projects, prototyping, and automation because of its simplicity show in figure 5.1. and versatility.

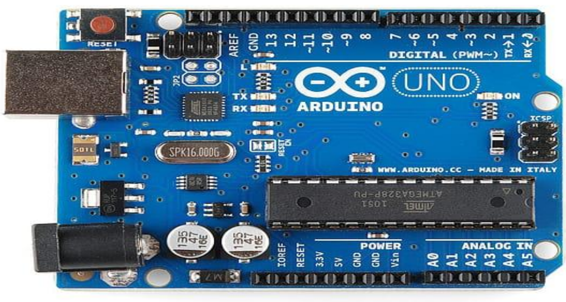


**Figure 5.1. Arduino Uno**

### 5.2.2 Light Dependent Resistors (LDRs)

A Light Dependent Resistor (LDR), also known as a photoresistor, is a sensor whose resistance changes based on the amount of light falling on it show in figure 5.2. It is commonly used to detect light intensity in electronic projects.

**Figure 5.2. Light Dependent Resistors**

### 5.2.3 Servo Motors

A Servo Motor is a rotary actuator that allows precise control of angular position, velocity, and acceleration. It consists of a motor coupled with a sensor for position feedback show in figure 5.3. and a control circuit. Servo motors are widely used in robotics, automation, and tracking systems.

- Operates on DC voltage (typically 4.8–6V for small servos).
- Can rotate to a specific angle (usually 0°–180°).
- Provides precise position control.

**Figure 5.3.Servo Motors**

### 5.2.4 Solar Panels

A solar panel is a device that converts sunlight into electrical energy using the photovoltaic effect. It is a key component show in figure 5.4.in solar power systems, used for generating clean

**Figure 5.4. Solar Panels**

### 5.2.5 Jumper Wires:

Jumper wires are short, flexible electrical wires used to make connections between components on a breadboard or between a breadboard and other devices like microcontrollers and sensors.They are commonly used in prototyping show in figure 5.5.

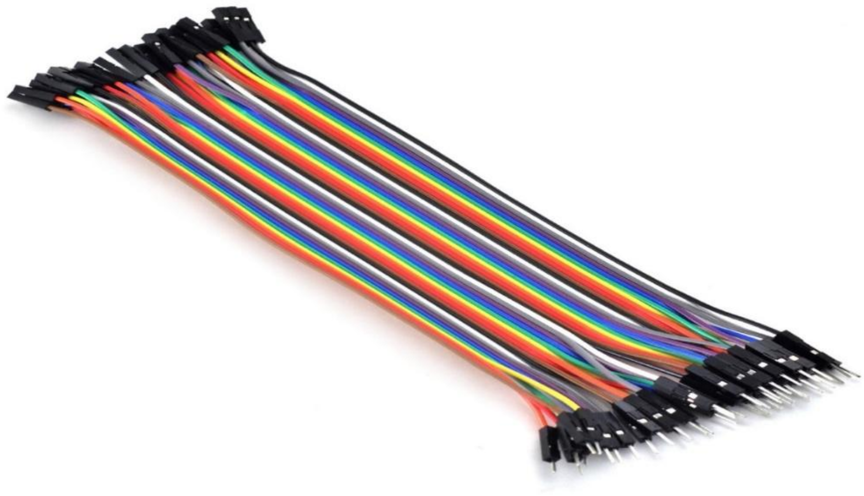

**Figure 5.5.Jumper Wires**

### 5.2.6 Breadboard / PCB

Breadboard is a solderless platform used to build and test electronic circuits quickly. A PCB (Printed Circuit Board) is a board with etched pathways that permanently show in figure 5.6. connect electronic components for a more durable and compact circuit.

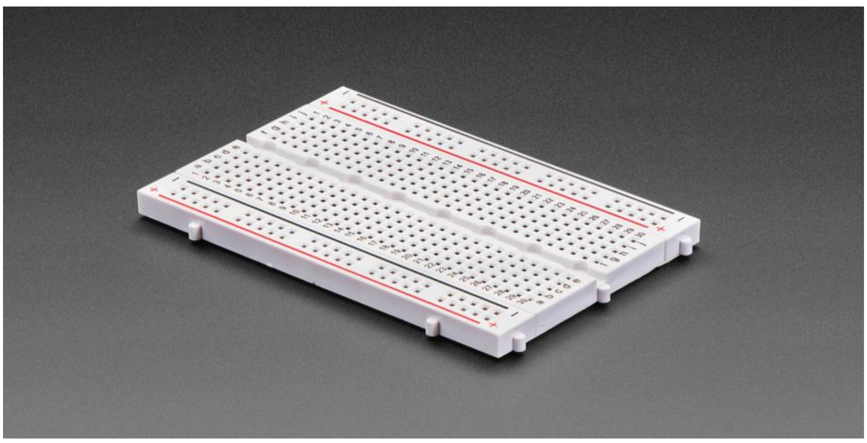

**Figure 5.6.Breadboard / PCB**

### 5.2.7 Resistors

A resistor is an electronic component that limits or regulates the flow of electric current in a circuit. show in figure 5.7. It is one of the most commonly used components in electronic circuits.


**Figure 5.7. Resistors**

- Mounting Frame – Holds the solar panels and allows movement according to the motors.
- Diodes / Transistors (Optional) – To protect circuits and control higher current if required.
- Mobile App or LCD Display (Optional) – For monitoring system status, sunlight intensity, and panel orientation.

Note: The choice of components depends on the size of the solar panels, power requirements, and desired system complexity. Using cost-effective and readily available components ensures easy replication and maintenance

## BLOCK DIAGRAM

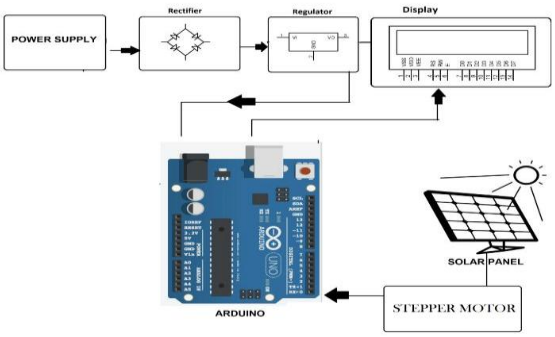


**Figure 5.8. BLOCK DIAGRAM**

## 5.3 THEORETICAL ANALYSIS

The theoretical analysis of a Sunlight Tracking System involves understanding how the system maximizes solar energy capture by aligning solar panels perpendicular to the sun's rays throughout the day.

### Principle of Operation

Solar panels generate maximum power when sunlight falls directly on their surface. The intensity of sunlight on the panel decreases if the angle of incidence deviates from 90°. By continuously adjusting the panel angle using sensors and motors, the system ensures optimal energy absorption.

### Sensor Functionality

**Light Dependent Resistors (LDRs):** Detect sunlight intensity.

**Working Principle:** LDRs change resistance based on the intensity of light falling on them. The Arduino reads these resistance changes as voltage levels to determine which direction receives more sunlight.

## Motor Control

**Servo Motors:** Rotate the panel along two axes (azimuth and elevation) to follow the sun.

**Control Algorithm:** The Arduino compares readings from multiple LDRs. It sends signals to the servo motors to move the panel toward the direction of maximum light.

## Energy Efficiency Calculation

The energy captured by a solar panel is theoretically proportional to the cosine of the angle (θ) between the sunlight and the panel surface:

$$E = E_{max} \times \cos(\theta) \qquad \text{----------------- [1]}$$

Where:

$E$= Energy captured at a given angle

$E_{max}$= Maximum energy when panel is perpendicular to sunlight

$\theta$= Angle of deviation from perpendicular sunlight

By minimizing $\theta$through tracking, the system maximizes energy output.

## System Advantages

- Increases energy generation compared to static panels.
- Reduces energy losses due to suboptimal panel orientation.
- Enhances overall reliability and performance.

## Conclusion

The theoretical analysis confirms that a sunlight tracking system can significantly improve solar energy capture by continuously aligning panels to the sun's position using sensors, microcontrollers, and motors.

## 5.4 ADVANTAGES

- **Maximum Power Output:**

  The system keeps the solar panel perpendicular to sunlight, resulting in 30–45% increase in power generation compared to fixed panels.

- **Higher Efficiency Throughout the Day**

  Since the panel continuously adjusts its position, it maintains optimal exposure from sunrise to sunset

- **Better Utilization of Solar Energy**

  Ensures that even low-intensity or diffused light is captured effectively.

- **Automatic Operation**

  The system uses sensors and control circuits to track the sun without human intervention.

## 5.5 APPLICATIONS

- **Solar Power Plants**

  Large-scale PV installations use trackers to significantly boost overall energy production.

- **Solar Water Heaters**

  Helps the heating panel stay aligned with the sun for efficient thermal output.
- **Solar Street Lights**

  Ensures maximum charging during daytime, especially useful in cloudy areas.
- **Solar-Based Research Equipment**

  Used in laboratories and observatories where sunlight alignment is critical.
- **Agricultural and Rural Energy Systems**

  Used for irrigation pumps, farm lighting, and rural micro-grids.

## 5.6 RESULT

The Sunlight Tracking System was successfully developed and tested. The system was able to detect the direction of maximum sunlight using LDR sensors and automatically rotate the solar panel towards the brightest light source using a servo/stepper motor.During testing, the tracker consistently adjusted its position throughout the day, ensuring that the solar panel remained aligned with sunlight. show in figure 5.9.

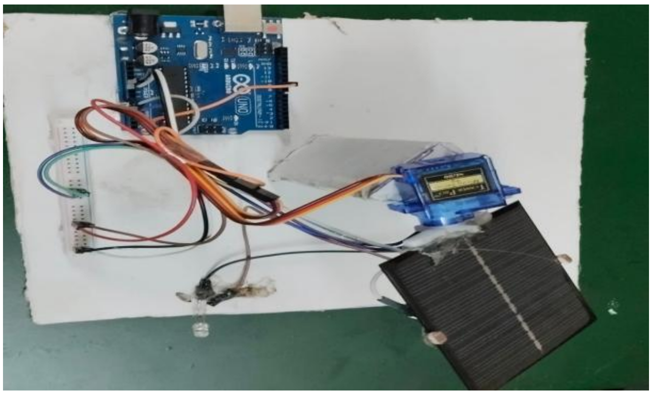

**Figure 5.9. RESULT**

# CHAPTER – 6

# CONCLUSION

The sunlight tracking system successfully illustrates how intelligent control and simple sensing techniques can enhance the overall efficiency of solar energy conversion. By using LDRs to detect the intensity difference between various directions, the system continuously adjusts the solar panel to maintain optimal alignment with the Sun. This dynamic tracking significantly increases the amount of solar radiation captured when compared to conventional fixed solar panels.

Through the integration of electronic components such as the microcontroller, motor driver, and servo or DC motor, the system operates smoothly, reliably, and with minimal power consumption. The project highlights the importance of automation in modern renewable energy solutions, emphasizing how even low-cost hardware can lead to notable improvements in performance.

Beyond demonstrating technical concepts, this project also reinforces the global need for sustainable energy technologies. As energy demands rise and environmental concerns grow, sunlight tracking systems provide a practical pathway toward maximizing clean energy production. Therefore, the project not only fulfills its technical objectives but also underlines its relevance for future green technology developments. It stands as a scalable, adaptable, and impactful solution for real-world solar applications.

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# APPENDIX

```
#include <Servo.h>
Servo servo;
int LDR_Left = A0;
int LDR_Right = A1;
int pos = 90;

void setup() {
  servo.attach(9);
  servo.write(pos);
  delay(1000);
  Serial.begin(9600);
}
void loop() {
  int leftValue = analogRead(LDR_Left);
  int rightValue = analogRead(LDR_Right);
  int diff = leftValue - rightValue;

  if (diff > 50 && pos < 180) {
    pos++;
    servo.write(pos);
    delay(10);
  } else if (diff < -50 && pos > 0) {
    pos--;
    servo.write(pos);
    delay(10);
  }

  Serial.print("Left: ");
  Serial.print(leftValue);
  Serial.print(" Right: ");
  Serial.println(rightValue);
}
```

# PHOTOGRAPHS

## 1.

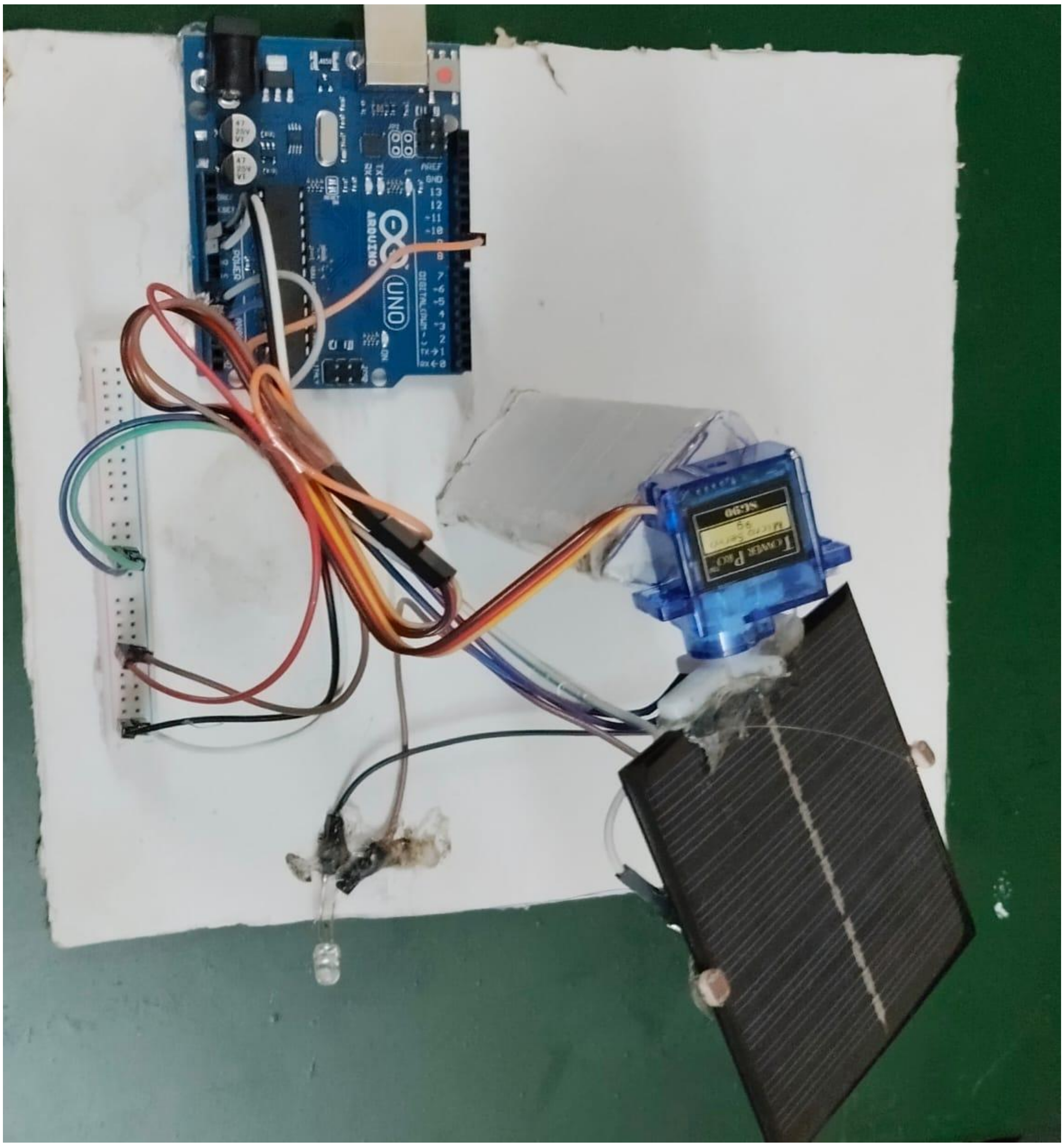


**Figure 5.10**

**2.**

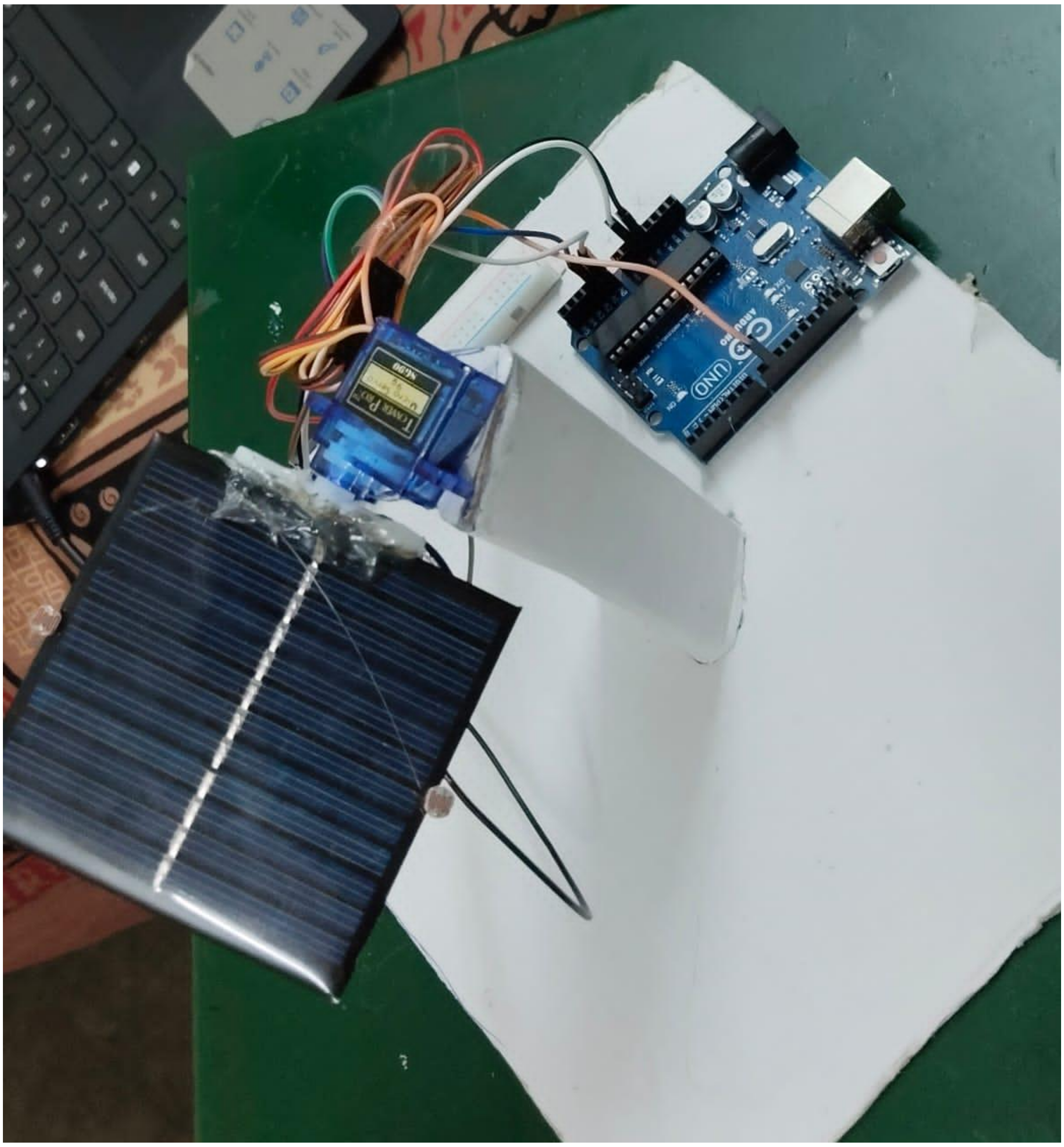

**Figure 5.11**